



Back-Up Lab (Core)

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Learning Objectives:

- Perform a backup utilizing software in Linux via the terminal.
- Schedule automated backups in Linux.

Overview

Depending on the Linux OS you are working with, there are different programs available to perform backups. Different Linux systems provide their own backup programs. The default program within Ubuntu, one of the most popular Linux distributions, is RSync. RSync works similarly to other backup products in that you can select the files and folders you would like to back up and specify the desired location of the backup. Utilize the resources available to you to gain an understanding of the RSync command. With RSync, files can easily be restored in the event they are lost.

Once you have verified the function of the RSync commands you created, you can use Crontab to schedule these backups on specified intervals. Again, utilize the resources available to you to gain an understanding of how to schedule the required Cron jobs.

Instructions

Use the instructions below to back up Kali Linux using Rsync:

1. To view the manual page, enter the command **rsync -m**. (As previously mentioned, RSync comes pre-installed in Kali Linux.)
2. Create a **daily** backup of the **/usr/share/** directory on the Desktop using Rsync and Crontab.
3. Create a **weekly** backup of the **/Documents/** directory on the Desktop using RSync and Crontab.
4. Include a screenshot of the scheduled cron jobs. These can be listed in the terminal with a crontab command.
5. Once you have uploaded your document with screenshots and explanations, you can disable the scheduled cron jobs.

Assignment Submission

Upload a single document containing a screenshot showing that you created a daily backup and another screenshot showing that you created a weekly backup. Be sure to provide an explanation of at least a sentence or two describing what each screenshot represents. Assignments without explanations for each screenshot will be returned for completion and resubmission.



Assignment Solution:

Create a daily backup of the `/usr/share/` directory on the *Desktop* using Rsync and Crontab:

- First, confirm content of the destination directory

```
(kali㉿kali)-[/usr/share]
$ cd /home/kali/Desktop
(kali㉿kali)-[~/Desktop]
$ pwd
/home/kali/Desktop
(kali㉿kali)-[~/Desktop]
$ ls -lrt
total 0
```

The screenshot indicates that nothing exists in the Desktop directory, currently.

- For starters, Initiate the backup of the `/usr/share/` directory on the Desktop using Rsync:

`rsync -av /usr/share/ /home/kali/Desktop/RSyncDaily`

This would create the **RSyncDaily** directory in Desktop, and backup the entire content of `/usr/share/` into it.

- a (archive): This option is a combination of several options that ensure the backup preserves permissions, ownership, timestamps, and recursively copies directories.
- v (verbose): Enables verbose mode, providing detailed information about the files being copied.

```
(kali㉿kali)-[~]
$ rsync -av /usr/share/ /home/kali/Desktop/RSyncDaily
sending incremental file list
created directory /home/kali/Desktop/RSyncDaily
./
hashcat-utils → ../lib/hashcat-utils
nikto → /var/lib/nikto
windows-binaries → windows-resources/binaries
GConf/
GConf/gsettings/
GConf/gsettings/gsettings-desktop-schemas.convert
GConf/gsettings/gvfs-dns-sd.convert
```

```
sent 4,084,168,894 bytes received 4,675,417 bytes 9,036,120.02 bytes/sec
total size is 4,066,307,418 speedup is 0.99
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1338) [sender=3.2.7]
```

```
(kali㉿kali)-[~]
$
```

- Confirm if the new folder/directory has been created:

```

(kali㉿kali)-[~/Desktop]
$ pwd
/home/kali/Desktop

(kali㉿kali)-[~/Desktop]
$ ls -all
total 20
drwxr-xr-x  3 kali kali  4096 Dec 29 14:45 .
drwx----- 20 kali kali  4096 Dec 28 23:39 ..
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily

(kali㉿kali)-[~/Desktop]
$ ls -lrt
total 12
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily

(kali㉿kali)-[~/Desktop]
$ ls -a
.  ..  RSyncDaily

(kali㉿kali)-[~/Desktop]
$ ls --all
.  ..  RSyncDaily

(kali㉿kali)-[~/Desktop]
$ ls -all
total 20
drwxr-xr-x  3 kali kali  4096 Dec 29 14:45 .
drwx----- 20 kali kali  4096 Dec 28 23:39 ..
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily

(kali㉿kali)-[~/Desktop]
$ ls -la
total 20
drwxr-xr-x  3 kali kali  4096 Dec 29 14:45 .
drwx----- 20 kali kali  4096 Dec 28 23:39 ..
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily

(kali㉿kali)-[~/Desktop]
$ █

```

Confirmed, as depicted in the screenshot above.

- Check the directory to confirm if the backup was done:

```
(kali㉿kali)-[~]
$ cd /home/kali/Desktop/RSyncDaily

(kali㉿kali)-[~/Desktop/RSyncDaily]
$ ls -lrt
total 1548
lrwxrwxrwx 1 kali kali 20 Jun 29 2020 hashcat-utils → ../lib/hashcat-utils
lrwxrwxrwx 1 kali kali 14 Feb 23 2023 nikto → /var/lib/nikto
lrwxrwxrwx 1 kali kali 26 Mar 3 2023 windows-binaries → windows-resources/binaries
drwxr-xr-x 3 kali kali 4096 Jun 17 2023 gdb
drwxr-xr-x 3 kali kali 4096 Jun 17 2023 gcc
drwxr-xr-x 3 kali kali 4096 Aug 11 12:52 util-linux
drwxr-xr-x 4 kali kali 4096 Aug 21 14:50 lintian
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 common-licenses
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 base-files
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 debconf
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 keyrings
drwxr-xr-x 4 kali kali 4096 Aug 21 14:51 nano
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 readline
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 sensible-utils
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 perl
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 cmake
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 ca-certificates
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 groff
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 man-db
drwxr-xr-x 44 kali kali 4096 Aug 21 14:51 terminfo
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 appopt
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 ucf
drwxr-xr-x 7 kali kali 4096 Aug 21 14:51 initramfs-tools
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 glib-2.0
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 fonts-droid-fallback
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 fontconfig
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 runit
drwxr-xr-x 8 kali kali 4096 Aug 21 14:51 fonts
drwxr-xr-x 6 kali kali 4096 Aug 21 14:51 poppler
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 iso-codes
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 gst-plugins-base
drwxr-xr-x 3 kali kali 4096 Aug 21 14:51 gettext
drwxr-xr-x 2 kali kali 4096 Aug 21 14:51 libthai
drwxr-xr-x 3 kali kali 4096 Aug 21 14:52 pulseaudio
```

Create a weekly backup of the **/Documents/** directory on the Desktop using RSync and Crontab.

- **/Documents/** directory indicated in the assignment does not exist.

```
(kali㉿kali)-[~/Documents]
$ cd /Documents
cd: no such file or directory: /Documents
```

- **/home/kali/Documents** exist, so this would be used for the weekly backup.

```
(kali㉿kali)-[~/Desktop]
$ cd /home/kali/Documents

(kali㉿kali)-[~/Documents]
$ ls -lrt
total 0
```

```

(kali㉿kali)-[~]
$ pwd
/home/kali
(kali㉿kali)-[~]
$ rsync -a /home/kali/Documents/ /home/kali/Desktop/RSyncWeekly
(kali㉿kali)-[~]
$ rsync -av /home/kali/Documents/ /home/kali/Desktop/RSyncWeekly
sending incremental file list
sent 52 bytes received 12 bytes 128.00 bytes/sec
total size is 0 speedup is 0.00
(kali㉿kali)-[~]
$

```

The screenshot above depicts the backup of the Documents directory on the Desktop and should create the directory '**RSyncWeekly**' with the content of the backup.

- Confirm that the new folder was created on the Desktop and confirm that the content is same size as what was indicated when the command was successfully executed.

```

(kali㉿kali)-[~]
$ cd /home/kali/Desktop
(kali㉿kali)-[~/Desktop]
$ ls -lrt
total 16
drwxr-xr-x  2 kali kali  4096 Nov 11 05:45 RSyncWeekly
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily
(kali㉿kali)-[~/Desktop]
$ cd RSyncWeekly
(kali㉿kali)-[~/Desktop/RSyncWeekly]
$ ls -lrt
total 0
(kali㉿kali)-[~/Desktop/RSyncWeekly]
$

```

Confirmed, screenshot shows 0 files just as was indicated after the execution of the rsync command.

Now to scheduling these backups as daily & weekly routines:

To schedule this rsync command for daily backups, tools such as cron can be used on Unix-based systems.

1. Open the crontab file for editing: **crontab -e**
2. Add a line to schedule the **rsync** command.
3. Save and exit the editor.

Include a screenshot of the scheduled cron jobs. These can be listed in the terminal with a **crontab** command.

```
(kali㉿kali)-[/usr/share]
$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow command
35 23 * * * rsync -a /usr/share/ /home/kali/Desktop/RSyncDaily_$(date +%Y\%m\%d\_%\H\%M\%S)
35 23 * * 5 rsync -a /home/kali/Documents/ /home/kali/Desktop/RSyncWeekly_$(date +%Y\%m\%d\_%\H\%M\%S)

(kali㉿kali)-[/usr/share]
$
```


Daily Backup:

```
35 23 * * * rsync -a /usr/share/ /home/kali/Desktop/RSyncDaily_$(date +%Y%m%d_%H%M%S)
```

This crontab entry represents a scheduled task that runs the **rsync** command at a specific time every day to synchronize the contents of the **/usr/share/** directory to a destination directory on the desktop. The destination directory includes a timestamp to create a unique directory each time the task runs.

- **Schedule:**

- **35 23 * * ***: The cron schedule specifies when the task will run:
 - **35**: The minute field. In this case, **35** means the task will run at the 35th minute of the hour.
 - **23**: The hour field. **23** means the task will run at the 23rd hour, which is 11:00 PM in 24-hour format.
 - Day of the Month (*): The cron job will run every day of the month.
 - Month (*): The cron job will run every month.
 - Day of the Week (*): The cron job will run every day of the week. (The days of the week are represented numerically, starting from 0 for Sunday and ending with 6 for Saturday).

- **Command:**

- **rsync -a /usr/share/ /home/kali/Desktop/RSyncDaily_\$(date +%Y%m%d_%H%M%S)**
This is the command that will be executed with the following options and arguments.
 - **-a**: Archive mode. Preserves permissions, ownership, timestamps, and recursively copies directories
 - **/usr/share/**: Source directory. The contents of this directory will be copied
 - **/home/kali/Desktop/RSyncDaily_\$(date +%Y%m%d_%H%M%S)**: The destination directory, which includes a timestamp generated by the **date** command. This timestamp ensures that each backup creates a new directory with a unique name.

In summary, this cron job is scheduled to run **rsync** at 11:35 PM every day, copying the contents of **/usr/share/** to a new directory on the desktop with a timestamp in the format **YYYYMMDD_HHMMSS**. This helps in organizing and distinguishing backups based on the date and time they were created.

Weekly Backup:

```
35 23 * * 5 rsync -a /home/kali/Documents/ /home/kali/Desktop/RSyncWeekly_$(date +%Y%m%d_%H%M%S)
```

This crontab entry represents a scheduled task that runs the **rsync** command at a specific time every day to synchronize the contents of the **/home/kali/Documents/** directory to a destination directory on the desktop. The destination directory includes a timestamp to create a unique directory each time the task runs.

- **Schedule:**

- **35 23 * * 5**: The cron schedule specifies when the task will run:
 - **35**: The minute field. In this case, **35** means the task will run at the 35th minute of the hour.
 - **23**: The hour field. **23** means the task will run at the 23rd hour, which is 11:00 PM in 24-hour format.
 - Day of the Month (*): The cron job will run every day of the month.
 - Month (*): The cron job will run every month.
 - Day of the Week (5): The cron job will run every Friday. (The days of the week are represented numerically, starting from 0 for Sunday and ending with 6 for Saturday).

- **Command:**

- **rsync -a /home/kali/Documents/ /home/kali/Desktop/RSyncWeekly_\$(date +%Y%m%d_%H%M%S)**
This is the command that will be executed with the following options and arguments.
 - **-a**: Archive mode. Preserves permissions, ownership, timestamps, and recursively copies directories
 - **/home/kali/Documents/**: Source directory. The contents of this directory will be copied
 - **/home/kali/Desktop/RSyncWeekly_\$(date +%Y%m%d_%H%M%S)**:
The destination directory, which includes a timestamp generated by the **date** command. This timestamp ensures that each backup creates a new directory with a unique name.

In summary, this cron job is scheduled to run **rsync** at 11:35 PM every Friday, copying the contents of **/home/kali/Documents/** to a new directory on the desktop with a timestamp in the format **YYYYMMDD_HHMMSS**. This helps in organizing and distinguishing backups based on the date and time they were created.

At the scheduled time, the cron job is automatically executed.

- To confirm, check the destination directory. Find screenshots below:

```
(kali㉿kali)-[~/Desktop]
└─$ pwd
/home/kali/Desktop
(kali㉿kali)-[~/Desktop]
└─$ ls -lrt
total 32
drwxr-xr-x  2 kali kali  4096 Nov 11 05:45 RSyncWeekly_20231229_233501
drwxr-xr-x  2 kali kali  4096 Nov 11 05:45 RSyncWeekly
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily_20231229_233501
drwxr-xr-x 352 kali kali 12288 Dec  3 03:38 RSyncDaily
```

Once you have uploaded your document with screenshots and explanations, you can disable the scheduled cron jobs:

- Disable the cron jobs & confirm that there are no more cron jobs available:

```
(kali㉿kali)-[~/Desktop]
└─$ crontab -r

(kali㉿kali)-[~/Desktop]
└─$ crontab -l
no crontab for kali

(kali㉿kali)-[~/Desktop]
└─$
```